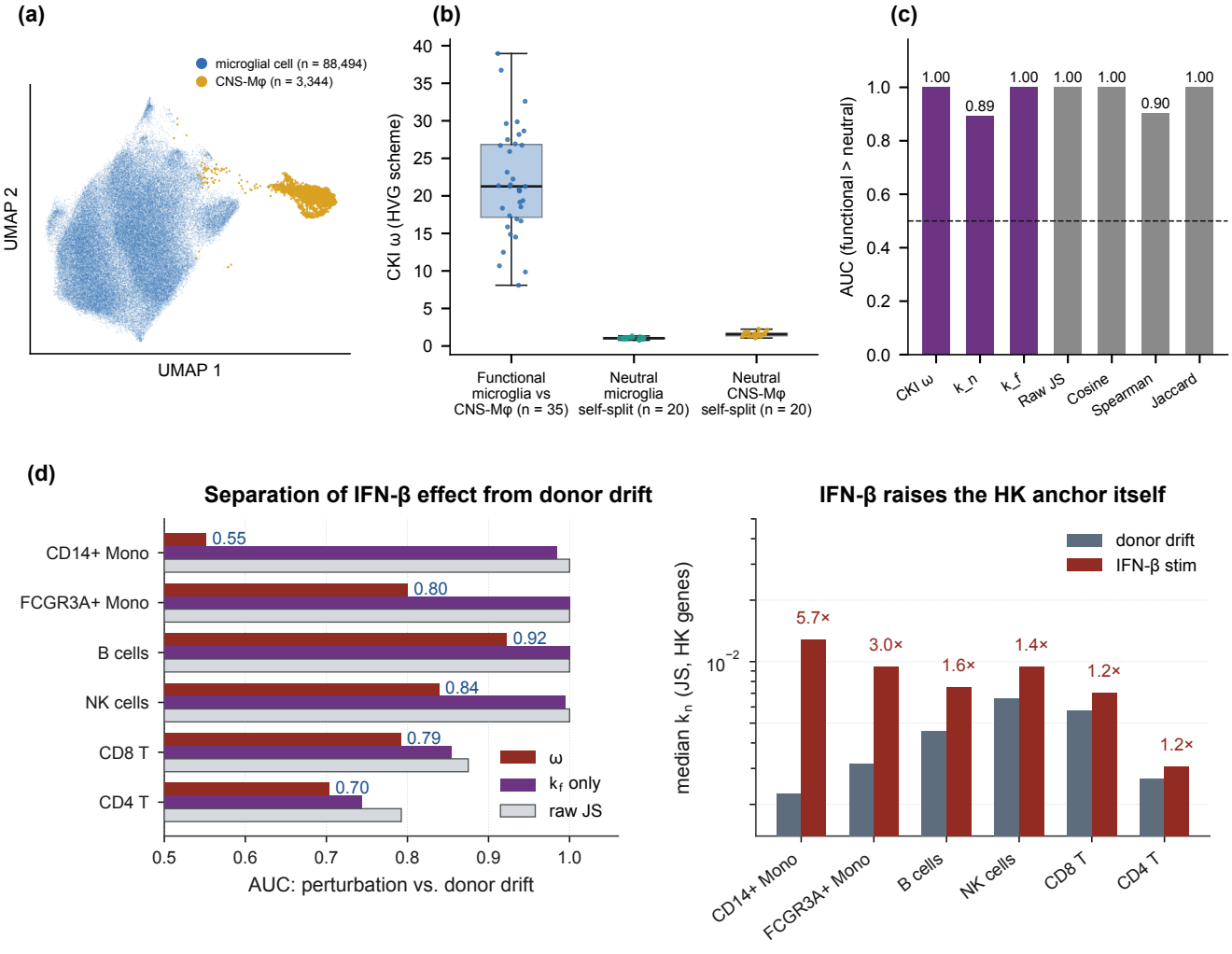

(a)
microglial cell (n = 88,494)
CNS-Mφ (n = 3,344)
UMAP 2
UMAP 1
(b)
CKI ω (HVG scheme)
Functional microglia vs CNS-Mφ (n = 35)
Neutral microglia self-split (n = 20)
Neutral CNS-Mφ self-split (n = 20)
(c)
AUC (functional > neutral)
1.00
0.89
1.00
1.00
1.00
0.90
1.00
CKI ω
k_n
k_f
Raw JS
Cosine
Spearman
Jaccard
(d)
Separation of IFN-β effect from donor drift
CD14+ Mono
FCGR3A+ Mono
B cells
NK cells
CD8 T
CD4 T
0.55
0.80
0.92
0.84
0.79
0.70
ω
k_f only
raw JS
AUC: perturbation vs. donor drift
IFN-β raises the HK anchor itself
median k_n (JS, HK genes)
donor drift
IFN-β stim
5.7×
3.0×
1.6×
1.4×
1.2×
1.2×